

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 4

Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objectives of this experiment are

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune pretrained CNN models.
- Compare classification performance of different architectures.

2. Learning Outcomes

After completing this experiment students will be able to

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3. Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision applications because they automatically learn hierarchical features from images. The evolution of CNNs has focused on improving classification accuracy while reducing computational complexity and training difficulties.

The progression of CNN architectures began with LeNet-5 and has evolved through AlexNet, VGGNet, GoogleNet, and ResNet. Each architecture introduced important innovations that significantly improved image classification performance.

4. Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

5. LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers. It demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction.

Advantages

- Small number of parameters
- Fast training
- Suitable for OCR

6. AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing the classification error significantly. It introduced ReLU activation, Dropout regularization and GPU training.

Major innovations

- ReLU activation
- Dropout
- GPU implementation
- Data augmentation

7. VGG16

VGG16 demonstrated that using multiple small 3×3 convolution filters could outperform larger filters while increasing network depth.

Advantages

- Excellent feature extraction
- Simple architecture
- High classification accuracy

8. Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogLeNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

9. GoogLeNet (Inception Network)

GoogLeNet, proposed by Szegedy *et al.* in 2014, introduced the **Inception Module**, which performs multiple convolution operations in parallel using different filter sizes. Instead of using only a single kernel size, GoogLeNet simultaneously applies 1×1 , 3×3 , 5×5 convolutions and max pooling. The outputs are concatenated to obtain multi-scale feature representations.

The use of 1×1 convolutions before larger kernels reduces the number of trainable parameters and computational complexity.

Advantages

- Multi-scale feature extraction.
- Fewer parameters than VGG16.
- High computational efficiency.
- Better classification performance.

10. ResNet

Increasing CNN depth often causes the **vanishing gradient problem**, making optimization difficult. ResNet solves this issue using **skip (residual) connections**, allowing gradients to flow directly through the network.

Instead of learning

$$H(x)$$

ResNet learns

$$F(x) = H(x) - x$$

thus,

$$Output = F(x) + x$$

where

- $F(x)$ = Residual Mapping
- x = Identity Mapping

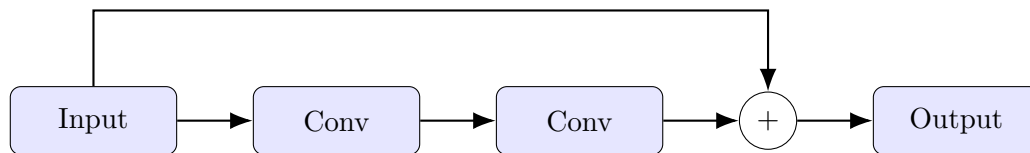


Figure 1: Residual Block in ResNet

Advantages

- Eliminates vanishing gradients.
- Enables very deep neural networks.
- Faster convergence.
- High ImageNet accuracy.

11. Dilated Convolution

Dilated convolution (also called atrous convolution) increases the receptive field without increasing the number of parameters.

The dilation rate determines the spacing between kernel elements.

For a standard convolution,

$$D = 1$$

For dilated convolution,

$$D > 1$$

Example

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{bmatrix}$$

where zeros indicate inserted gaps.

Applications

- Semantic segmentation
- Medical image analysis

- Satellite imagery
- Object localization

12. Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps. Unlike pooling, which reduces image size, transpose convolution performs learnable upsampling.

Applications

- Image Super Resolution
- Autoencoders
- GANs
- Image Generation
- Semantic Segmentation

13. Transfer Learning

Transfer Learning utilizes a pretrained CNN model that has already learned general image features from a large dataset such as ImageNet.

Instead of training from scratch,

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

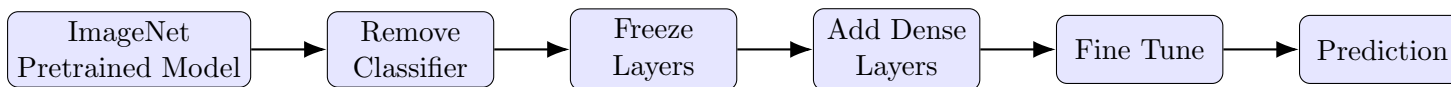


Figure 2: Transfer Learning Workflow

Advantages

- Faster convergence.
- Higher accuracy on small datasets.
- Reduced training time.
- Lower computational cost.

14. Dataset

Dataset: CIFAR-10

- Training Images : 50,000
- Testing Images : 10,000
- Number of Classes : 10
- Image Size : $32 \times 32 \times 3$
- Classes: Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck.

15. Experimental Procedure

Task 1: Dataset Preparation

1. Load the CIFAR-10 dataset using TensorFlow/Keras.
2. Normalize the pixel values to the range $[0,1]$.
3. Display ten sample images.

4. Print the dimensions of the training and testing datasets.

Task 2: Transfer Learning

Load any one of the following pretrained CNN models.

- VGG16
- ResNet50
- MobileNetV2
- EfficientNetB0 (Optional)

Perform the following steps.

1. Load pretrained ImageNet weights.
2. Remove the original classification layer.
3. Freeze the convolutional base.
4. Add a Global Average Pooling layer.
5. Add a Dense layer with ReLU activation.
6. Add the output layer with Softmax activation.

Task 3: Model Training

Train the model using

Optimizer	Adam
Learning Rate	0.001
Batch Size	32
Epochs	10–20
Loss Function	Categorical Cross Entropy
Evaluation Metric	Accuracy

Task 4: Fine Tuning

1. Unfreeze the last convolution block.
2. Train for another 5–10 epochs.
3. Compare the classification accuracy before and after fine tuning.

Task 5: Model Evaluation

Evaluate the trained model using

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- Classification Report

16. Hyperparameter Study

Compare the performance using different settings.

Hyperparameter	Values
Learning Rate	0.001, 0.0001
Batch Size	16, 32, 64
Epochs	10, 20
Optimizer	Adam, SGD
Dense Units	128, 256
Frozen Layers	All, Partial

17. Source Code

The complete source code is attached separately in the GitHub repository and submitted along with this report.

GitHub Repository:

<https://github.com/Sadhana2006-art/Deep-Learning-Lab>

18. Plots with Inference

18.1 Sample Images



Figure 1: Sample Images from the Dataset

Interpretation:

The picture has ten sample pictures from the CIFAR-10 dataset. These pictures show things like Frog, Truck, Deer, Automobile, Bird, Horse, Ship and Cat. Each picture is a picture, with 32×32 pixels.

18.2 Training and Validation Accuracy

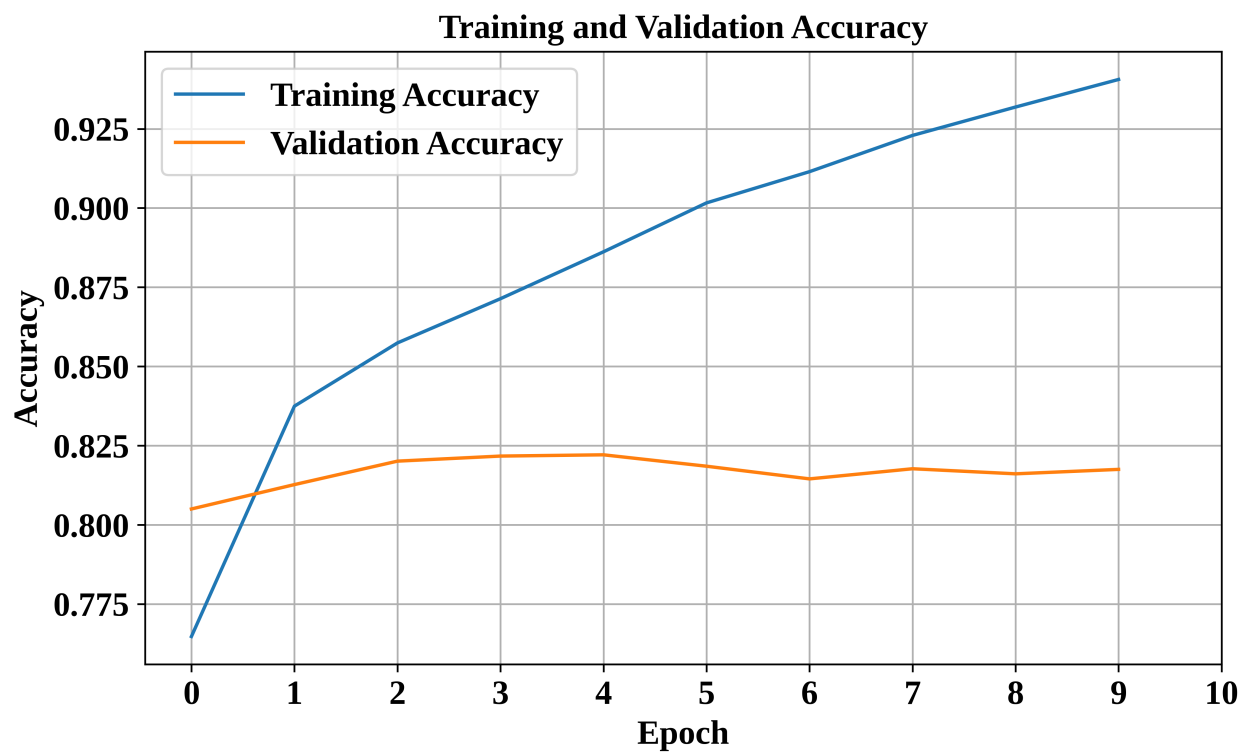


Figure 2: Training and Validation Accuracy

Interpretation:

There was steady improvement in training accuracy from about 76.5% to 94.0%, showing that the model has learned well. But there were no improvements in validation accuracy, which was at 81-82% even after the first few epochs. This is a case of overfitting.

18.3 Training and Validation Loss

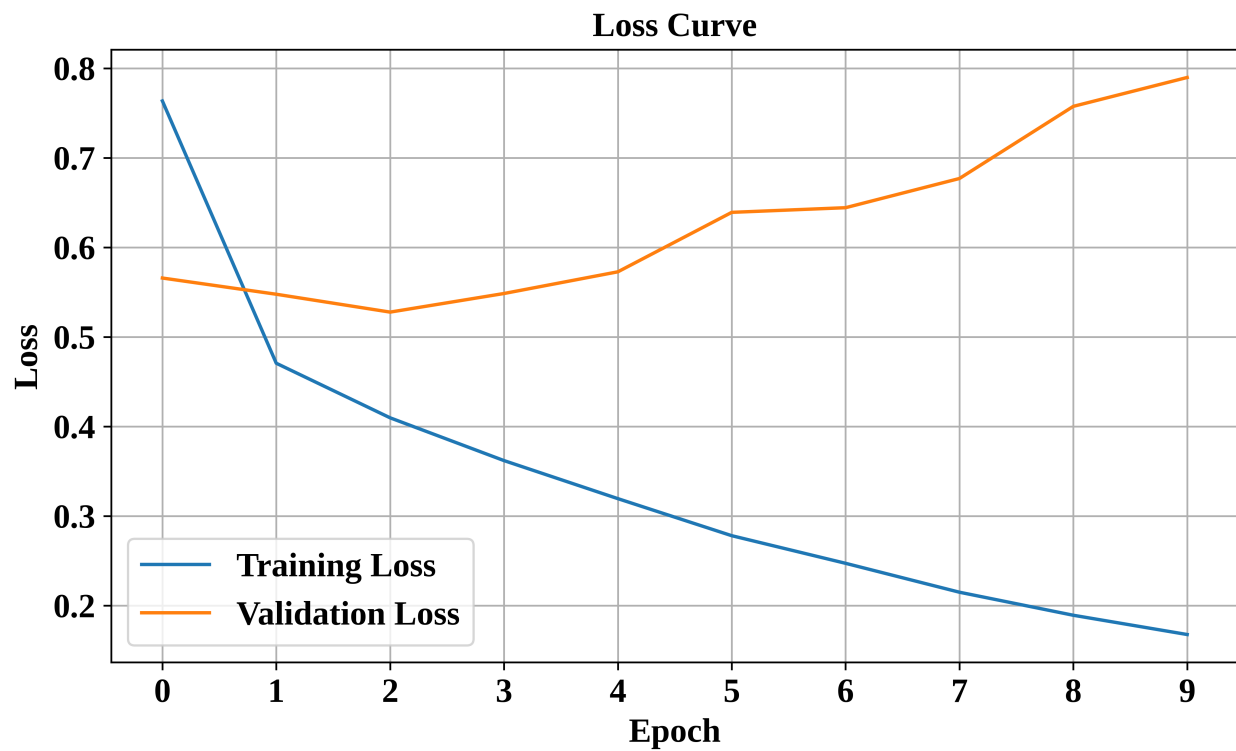


Figure 3: Training and Validation Loss

Interpretation:

Training loss kept decreasing from around 0.76 to 0.17, indicating optimization of the model. On the other hand, the validation loss kept on decreasing but began increasing after the early epochs, suggesting that the model is overfitting to the training data set.

18.4 Confusion Matrix

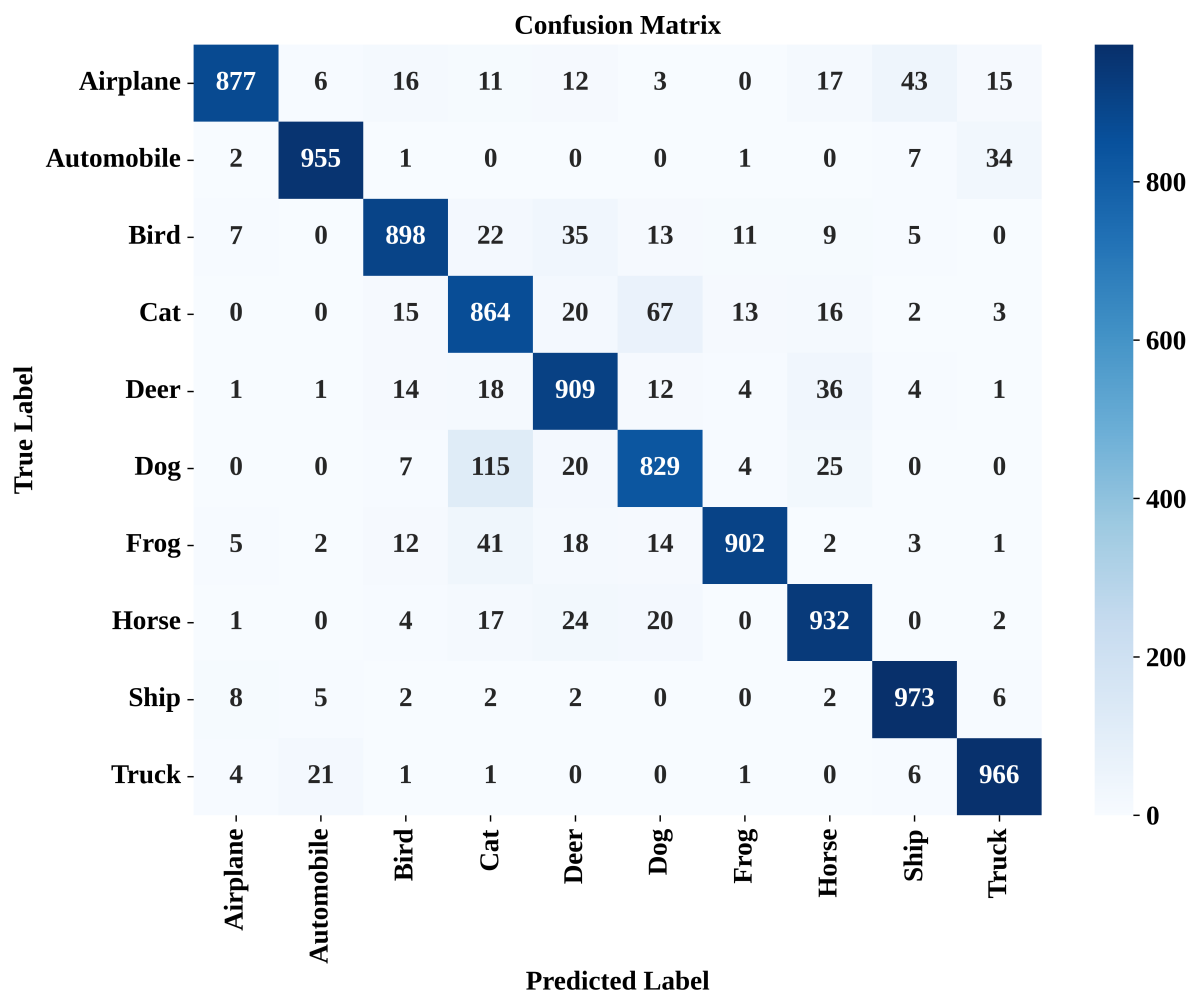


Figure 4: Confusion Matrix

Interpretation:

Confusion matrix shows that the model is capable of producing good accuracy because of the presence of correct predictions for most samples on the diagonal. The classes Ship (973), Truck (966), and Automobile (955) show very good recognition accuracy, whereas there is some confusion among visually close classes like Cat and Dog.

18.5 Misclassified Images

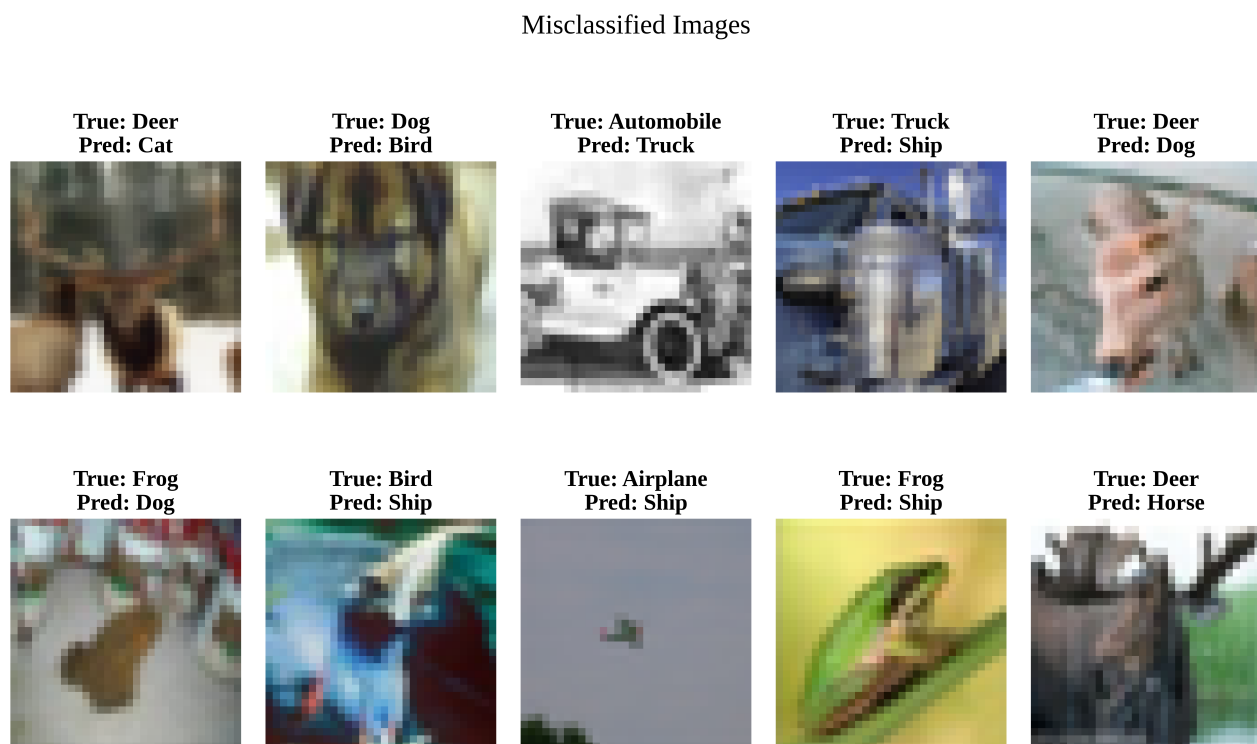


Figure 5: Misclassified Images

Interpretation:

The misclassified pictures prove that the network is having trouble recognizing images of classes with similar visual properties, like Deer-Cat, Deer-Dog, and Automobile-Truck. The factors which affect misclassification are poor resolution of the image, background interference, and similarity of objects.

19. Results

19.1 Performance Metrics

Metric	Value
Training Accuracy	94.06%
Validation Accuracy	91.05%
Precision	91.24%
Recall	91.05%
F1-score	91.08%
Training Time	769.27 seconds
Total Parameters	14,781,642

19.2 Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time (s)
LeNet-5	83,126	54.45	77.5
AlexNet	36,925,322	71.19	343.1
GoogleNet	22,066,346	69.58	296.1
ResNet50	23,851,274	86.38	392.7

20. Discussion Questions

1. **Why is AlexNet considered a breakthrough in deep learning?**

AlexNet has been regarded as a breakthrough because it has proven that deep CNNs can be very effective in performing large-scale image classification. This model uses several convolutional layers, Rectified Linear Unit (ReLU), dropout, and GPU training, which enhance the performance of deep learning models.

2. **Why does VGG16 use only 3×3 convolution filters?**

The reason behind the use of 3×3 convolutional filters by VGG16 is that by combining smaller filters into a stack, a larger receptive field is achieved using a fewer number of parameters compared to one large filter.

3. **Explain the advantages of the Inception module.**

The Inception module utilizes convolutions with various filter dimensions, such as 1×1 , 3×3 , and 5×5 , in addition to pooling. This way, the network is able to extract features of varying spatial dimensions. The 1×1 convolutions further serve the purpose of minimizing computational load and number of parameters.

4. **What is the purpose of residual learning?**

The reason behind the use of residual learning is that it makes the training of very deep neural networks much more convenient. In this case, instead of training the model for direct learning of the function or mapping, residual learning is used to learn the difference between the input and output in each block.

5. **Differentiate LeNet and ResNet.**

LeNet is one of the earliest and shallow CNNs that were developed to recognize handwritten digits. The network consists of a few convolutional and pooling layers.

ResNet is one of the deep CNNs which utilizes residual blocks with shortcuts. These enable ResNet to build very deep networks without the issues such as vanishing gradients.

6. **What is Transfer Learning?**

Transfer Learning is a concept in which the knowledge gained from a model in one particular dataset or problem is transferred to another problem. For instance, a CNN can be trained using a large dataset such as the ImageNet dataset, and this will act as a feature extractor in another image classification problem.

7. **Why is fine tuning required?**

It is essential to fine-tune the model for making necessary adjustments based on the new data set. Initially, the classification layers are trained, and then some of the pretrained layers can be made trainable by applying a low learning rate.

8. **Explain the difference between dilated convolution and transpose convolution.**

With dilated convolution, there will be spaces left between the elements of the kernel, enabling the convolution operation to span a larger receptive field by not requiring an increase in kernel parameters.

A transpose convolution is applied to upscale the spatial dimensions of the feature maps. It is applied in areas like image generation, segmentation, and decoding networks.

9. **Why do pretrained models converge faster?**

Training of pretrained models is fast since it already has meaningful features derived from a huge amount of data. The model does not have to learn from scratch features like edges and texture; instead, it uses pre-existing meaningful weights and adjusts them according to the task at hand.

10. Compare the computational complexity of LeNet and ResNet.

LeNet has an architecture that is comparatively simple, since it has fewer layers and parameters; hence, it uses less computational power and takes less time to train. ResNet has numerous layers and parameters, and thus, it needs more computational resources.

Nevertheless, due to the existence of the residual connections in ResNet, it is possible to have a deep network which is effectively trained. Therefore, LeNet is computationally cheaper, whereas ResNet offers more representational power.

21. Additional Exercises

1. Implement transfer learning using VGG16.
2. Repeat the experiment using ResNet50.
3. Compare Adam and SGD optimizers.
4. Compare training with frozen layers and fine tuning.
5. Evaluate the model using Precision, Recall and F1-score.
6. Compare the performance of LeNet, AlexNet and ResNet.

22. References

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8. Keras Documentation: <https://keras.io>